

Regular elements of algebraic groups and of Lie algebras

SGA 3, Exposé XIII

English draft

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About this draft

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Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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Exposé XIII. Regular elements of algebraic groups and of Lie algebras

by A. Grothendieck

1. An auxiliary lemma on varieties with operators

Let S be a prescheme, G an S -group prescheme acting on the left on an S -prescheme V , W a closed sub- S -prescheme of V , and N its stabilizer in G , the subgroup of G whose points with values in an S' over S are the $g \in G(S')$ such that $g \cdot W_{S'} = W_{S'}$. One equips $((\text{Sch})/S)$ with the faithfully flat quasi-compact topology, and identifies G, V, W with the corresponding sheaves (cf. IV). One shall therefore reason in the category of sheaves on $((\text{Sch})/S)$, and in this section the expression “locally” refers to the topology just specified on $((\text{Sch})/S)$. Note that N is a sheaf; consider the quotient sheaf G/N . One sees immediately that it is isomorphic to the following functor: to each S' over S , one associates the set of subsheaves W' of $V_{S'}$ which are locally conjugate to $W_{S'}$ under the group G . Let X be the subsheaf of $G/N \times_S V$ whose value, for every S' over S , is the set of (W', v) , where W' is as above and v is a section of W' over S' (thus a section of $V_{S'}$ over S'). Let Z be the inverse image of X in $G \times_S V$, so that one has the cartesian diagram

$$(x) \quad \begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow & & \downarrow i \\ G \times_S V & \longrightarrow & G/N \times_S V, \end{array}$$

where i is the canonical immersion, and the second horizontal arrow comes from the canonical morphism $G \rightarrow G/N$. Since this associates to the point $g \in G(S')$ the subsheaf $g \cdot W_{S'}$ of $V_{S'}$, one sees that $Z(S')$ is the set of pairs $(g, v) \in G(S') \times V(S')$ such that $v \in g \cdot W_{S'}(S')$. Consequently, Z is isomorphic to the sheaf $G \times_S W$, by means of the isomorphism

$$G \times_S W \xrightarrow{\sim} Z$$

defined by $(g, w) \mapsto (g, g \cdot w)$. Thus the preceding cartesian diagram gives the cartesian diagram

$$(xx) \quad \begin{array}{ccc} G \times_S W & \xrightarrow{q} & X \\ \lambda \downarrow & & \downarrow i \\ G \times_S V & \longrightarrow & G/N \times_S V \end{array}$$

where $\lambda(g, w) = (g, g \cdot w)$, hence $q(g, w) = (\bar{g}, g \cdot w)$, where $\bar{g}$ denotes the image of g under the canonical map $G(S') \rightarrow (G/N)(S')$. Finally, one sees from diagram (x) that $Z \rightarrow X$ makes Z a principal bundle with base X and group N acting on the right by $(g, v) \cdot n = (gn, v)$, so that in (xx), $q : G \times_S W \rightarrow X$ makes $G \times_S W$ a principal bundle with base X and group N acting on the right by

$$(g, w) \cdot n = (gn, n^{-1} \cdot w).$$

One summarizes the main preceding morphisms in the following diagram:

$$(D) \quad \begin{array}{ccccc} & & G \times_S W & & \\ & & \downarrow q & \searrow \varphi & \\ G/N & \xleftarrow{p} & X & \xrightarrow{\psi} & V \\ & \swarrow \text{pr}_1 & \downarrow i & \searrow \text{pr}_2 & \\ & & G/N \times_S V & & \end{array}$$

where $\psi = \text{pr}_2 \circ i$ and $\varphi = \psi \circ q$, i.e. $\varphi(g, w) = g \cdot w$. If v is a section of V over S , the subsheaf X_v of X which is the inverse image of this section under ψ is given by $X_v(S') =$ the set of subsheaves W' of $V_{S'}$ which are locally conjugate to $W_{S'}$ under G , and which contain the section $v_{S'}$ of $V_{S'}$, while the subsheaf of $G \times_S W$ which is the inverse image of v under φ is isomorphic to the subsheaf M_v of G given by $M_v(S') =$ the set of $g \in G(S')$ such that $v_{S'} \in g \cdot W(S')$, i.e. such that $g^{-1}v_{S'} \in W(S')$. If v is a section of W and not merely of V , then M_v clearly contains N .

In these descriptions, one has not used the fact that G, V, W were representable (nor that the site on which one works is defined in terms of preschemes!). But suppose now that N is representable and faithfully flat and quasi-compact over S , and that G/N is representable. When S is the spectrum of a field, and G is of finite type over k , one knows that this hypothesis is necessarily satisfied (VIII 6 and VI_B.11.18). One then sees from the cartesian diagram (xx), using faithfully flat quasi-compact descent theory and the fact that $Z \rightarrow G \times_S V$ is a closed immersion, that X is representable (it is obtained by descent of the closed subscheme Z of $G \times_S V$ along the faithfully flat and quasi-compact morphism $G \times_S V \rightarrow G/N \times_S V$). Thus diagram (D) is a diagram of morphisms of preschemes over S .

From now on one assumes that S is the spectrum of a field k , and that G, V, W are of finite type over k . Let $\mathfrak{n}$ be the Lie algebra of N , so that $\dim N \leq \text{rank } \mathfrak{n}$, equality holding if and only if N is smooth over k (Exp VI⁽¹⁾). Let $a \in W(k)$, and consider the subscheme M_a of G defined above, containing N , and isomorphic to $\varphi^{-1}(a)$; one shall denote by $\mathfrak{m}_a$ its Zariski tangent space at the identity element e of N , so that one has

$$(1) \quad \mathfrak{n} \subset \mathfrak{m}_a, \quad \dim N \leq \text{rank } \mathfrak{n} \leq \mathfrak{m}_a.$$

Lemma 1.1. *With the preceding notation:*

a) *Consider the following conditions:*

- (i) $\mathfrak{n} = \mathfrak{m}_a$ and N is smooth over k .
- (i bis) $\dim N = \text{rank}_k \mathfrak{m}_a$.
- (ii) $\psi : X \rightarrow V$ is unramified at $(\bar{e}, a)$.
- (iii) M_a and N coincide in a neighborhood of e .

Then one has the implications (i) $\Leftrightarrow$ (i bis) $\Rightarrow$ (ii) $\Leftrightarrow$ (iii).

b) *Suppose that $\varphi : G \times_S W \rightarrow V$ is smooth at (e, a) . Then M_a is smooth over k at e , and ψ is smooth at $(\bar{e}, a)$.*

PROOF. a) The equivalence of (i) and (i bis) follows at once from relations (1) and from the fact noted above that N is smooth over k if and only if $\dim N = \text{rank}_k \mathfrak{n}$. On the other hand, consider the inclusion morphism $N \rightarrow M_a$; it is well known⁽²⁾ that if N is smooth over k at e and the tangent map at e is surjective, then $N \rightarrow M_a$ is smooth at e , and hence (being an immersion) is an isomorphism at e , which shows that (i) implies (iii). To prove the equivalence of (ii) and (iii), consider, as above, $X_a = \psi^{-1}(a)$ and use the isomorphism

⁽¹⁾**N.D.E.** Having been unable to identify this reference, we refer to Theorem II.5.2.1 of the book: M. Demazure & P. Gabriel, *Groupes algébriques I*, Masson (1970).

⁽²⁾**N.D.E.** See, for example, EGA IV₄, Th. 17.11.1 d).

$M_a \simeq \varphi^{-1}(a) \simeq q^{-1}(X_a)$ to obtain a morphism $p_a : M_a \rightarrow X_a$ which makes M_a a principal homogeneous bundle with group N_{X_a} . Consider the following diagram

$$\begin{array}{ccc} M_a & \xleftarrow{j_a} & N \\ p_a \downarrow & & \downarrow \\ X_a & \xrightarrow{j'_a} & S = \operatorname{Spec}(k), \end{array}$$

where $\operatorname{Spec}(k) \rightarrow X_n$ is defined by the point $(\bar{e}, a)$ of X , and $j_a : N \rightarrow M_a$ is the canonical immersion. Saying that ψ is unramified at $(\bar{e}, a)$ means that j'_a is an open immersion, or again that it induces an isomorphism $S \xrightarrow{\sim} \operatorname{Spec}(\mathcal{O}_{X_a, a})$. Since p_a is flat, it is equivalent to say that the morphism obtained from the preceding one by the base change $\operatorname{Spec}(\mathcal{O}_{M_a, e}) \rightarrow \operatorname{Spec}(\mathcal{O}_{X_a, b})$ is an isomorphism; now this resulting morphism is precisely the morphism $\operatorname{Spec}(\mathcal{O}_{N, e}) \rightarrow \operatorname{Spec}(\mathcal{O}_{M_a, e})$, which proves the equivalence of (ii) and (iii).

One will moreover note that the proof shows that conditions (ii), (iii) imply the following condition, apparently stronger than (iii):

(iii bis) N is an open and closed subscheme of M .

b) The first assertion follows from the fact that M_a is isomorphic to $\varphi^{-1}(a)$, the second from the fact that q is flat and $q(e, a) = (\bar{e}, a)$. $\square$

2. Density theorem and theory of regular points of G

One shall apply the constructions and notation of the preceding section in the case where G is a smooth connected algebraic group over k , where $V = G$ on which G acts by inner automorphisms, and where W is a smooth connected algebraic subgroup H of G . One shall denote by $\mathfrak{g}$ the Lie algebra of G , by $\mathfrak{h}$ that of H , by N the normalizer of H in G , and by $\mathfrak{n}$ the Lie algebra of N . If $a \in G(k)$, one shall again denote, as in no. 1, by M_a the inverse of its transporter into H , so that if $a \in H(k)$, one has $N \subset M_a$; in this case, one denotes by $\mathfrak{m}_a$ the Zariski tangent space of M_a at the identity element e of G . Note that

$$\mathfrak{h} \subset \mathfrak{n} \subset \mathfrak{m}_a \subset \mathfrak{g} \quad \text{for } a \in H(k).$$

One shall need the

Lemma 2.0. *For $H = N^0$ to hold, it is necessary and sufficient that $(\mathfrak{g}/\mathfrak{h})^H = 0$ (where the left-hand side denotes the subspace of invariants under the action of H deduced from the adjoint representation). When this condition is satisfied, N is smooth and one has $\dim X = \dim G$. In all cases, $\dim X \leq \dim G$, and this inequality is an equality if and only if H has finite index in N .*

Indeed, one has seen (II 5.2.3 (i)) that $\mathfrak{n}$ is equal to the inverse image of $(\mathfrak{g}/\mathfrak{h})^H$ under the morphism $\mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{h}$, hence $(\mathfrak{g}/\mathfrak{h})^H = 0$ is equivalent to $\mathfrak{h} = \mathfrak{n}$, which is also equivalent (H being a smooth connected algebraic subgroup of the algebraic group N) to $H = N^0$ (cf. VI.2). This clearly implies that N is smooth. On the other hand, one has

$$\dim X = (\dim G - \dim N) + \dim H = \dim G - (\dim N - \dim H),$$

so that one has

$$\dim X \leq \dim G,$$

equality being attained if and only if $\dim H = \dim N$, i.e. if and only if H has finite index in N . This is the case in particular if $H = N^0$, which completes the proof of 2.0.

Theorem 2.1. *Let G be a smooth connected algebraic group over the algebraically closed field k , H a smooth connected algebraic subgroup, N its normalizer, $\mathfrak{g}, \mathfrak{h}, \mathfrak{n}$ the Lie algebras, $X = G \times^N H$ the scheme (fibered over G/N with typical fiber H) introduced in no. 1, and $\psi : X \rightarrow G$ the canonical morphism (whose image is also the image of $\varphi : G \times H \rightarrow G$ defined by $\varphi(g, h) = \operatorname{int}(g)h = ghg^{-1}$). The following conditions are all equivalent:*

- (i) H contains a Cartan subgroup (XII 1) C of G .
- (i bis) H has the same reductive rank and the same nilpotent rank (XII 1) as G .
- (ii) H contains a maximal torus T of G , and $(\mathfrak{g}/\mathfrak{h})^T = 0$.
- (iii) The set of conjugates of H containing a given maximal torus is finite and nonempty, and H has finite index in its normalizer.
- (iv) There exists $a \in H(k)$ which is contained in only finitely many conjugates of H (or merely such that $\psi^{-1}(a)$ has an isolated point), and H has finite index in its normalizer.
- (iv bis) The morphism $\psi : X \rightarrow G$ is generically quasi-finite (i.e. there exists a dense open subset of X on which ψ is quasi-finite), and H has finite index in its normalizer.
- (v) There exists a dense open subset U of G such that for every $x \in U(k)$, the set of conjugates of H containing x is finite and nonempty, i.e. $\psi : X \rightarrow G$ is dominant and generically quasi-finite.
- (vi) There exists a dense open subset U of G such that every $x \in U(k)$ is contained in a conjugate of H , i.e. $\psi : X \rightarrow G$ is dominant.
- (vii) There exists $a \in H(k)$ such that the subspace of $\mathfrak{g}/\mathfrak{h}$ of fixed points of $\text{ad}_{\mathfrak{g}/\mathfrak{h}}(a)$ is zero.

Moreover, these conditions imply that H is its own connected normalizer, i.e. N is smooth and $\dim H = \dim N$, and that $\psi : X \rightarrow G$ is generically étale.

PROOF. By 2.0, one has $\dim X \leq \dim G$, with equality if and only if $\dim H = \dim N$, i.e. H has finite index in N . From the inequality $\dim X \leq \dim G$ it follows that ψ is dominant if and only if it is dominant and generically quasi-finite, or again if and only if ψ is generically quasi-finite and $\dim X = \dim G$. Since this last equality also means, by 2.0, that H has finite index in N , one has proved the equivalence of (vi), (v), (iv bis). The equivalence of (iv) and (iv bis) is immediate.

The equivalence of (i) and (i bis) is immediate from the definitions, and is left to the reader. On the other hand, if H contains a Cartan subgroup C of G , it contains the maximal torus T of C , which is a maximal torus of G . Since C is the centralizer of T , its Lie algebra $\mathfrak{c}$ is given by

$$\mathfrak{c} = \mathfrak{g}^T$$

(II, 5.2.3 (ii)). Thus, since $H \supset C$, whence $\mathfrak{h} \supset \mathfrak{n}$, it follows that $\mathfrak{h} \subset \mathfrak{g}^T$, which is equivalent, by I, 4.7.3, to the relation

$$(x) \quad (\mathfrak{g}/\mathfrak{h})^T = 0.$$

Conversely, suppose that H contains the maximal torus T and that the preceding relation holds, i.e. that one has $\mathfrak{h} \supset \mathfrak{n}$; I say that H contains the centralizer C of T (which will establish (i) $\Leftrightarrow$ (ii)). This follows from the

Lemma 2.1.1. *Let G be a smooth algebraic group over the field k , T a subgroup of multiplicative type of G , C its connected centralizer (equal to the centralizer of T if G is connected and T a torus, (XII 6.6 b)), and H a smooth subgroup of G containing T . For H to contain C , it is necessary and sufficient that its Lie algebra $\mathfrak{h}$ contain that $\mathfrak{c}$ of C .*

Indeed, one knows (XI 2.4) that $\text{Centr}_G(T)$ is smooth over k , hence C is smooth over k ; likewise $\text{Centr}_H(T)$ is smooth over k ; now $\text{Centr}_H(T) = \text{Centr}_G(T) \cap H$ has $\mathfrak{h} \cap \mathfrak{c}$ as Lie algebra, so the hypothesis implies that the smooth subgroup $\text{Centr}_H(T)$ of the smooth group $\text{Centr}_G(T)$ has the same Lie algebra, hence it contains the connected component C of the latter, so H contains C . Q.E.D.

Let us prove the equivalence of (i) and (iii), which amounts to proving that if H contains the maximal torus T of G , then the condition $H \supset C$ (which is also equivalent to (x) above, as one has just seen) is equivalent to the fact that H has finite index in its normalizer and that the set of conjugates of H containing T is finite. If H contains C , hence if one has (x),

then a fortiori

$$(xx) \quad (\mathfrak{g}/\mathfrak{h})^H = 0,$$

now one knows that $\mathfrak{n}$ is the inverse image of the left-hand side of the preceding relation under the canonical homomorphism $\mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{h}$ (II 5.2.3 (i)), so the preceding relation also means, by 2.0, that $H = N^0$; a fortiori H has finite index in its normalizer. Consider now the diagram of subgroups

$$\begin{array}{ccccc} T & \longrightarrow & N(T) \cap H & \longrightarrow & H \\ & & \downarrow & & \downarrow \\ & & N(T) \cap N(H) & \longrightarrow & N(H) \\ & & \downarrow & & \\ & & N(T) & & \end{array}$$

Using the conjugacy theorem for maximal tori in H (XII 6.6 a)), one sees that every conjugate of H containing T is conjugate to H by an element of $N(T)(k)$, hence that the set of conjugates of H containing T is in one-to-one correspondence with the set of points of $N(T)/N(T) \cap N(H)$ with coefficients in k ; now, since $H \supset C$, one has $N(T) \cap H \supset C$, so the preceding set is a quotient of $(N(T)/C)(k)$, which is a finite set, and is therefore finite. This proves that (i) $\Rightarrow$ (iii). Conversely, suppose (iii), i.e. $N(T)/N(T) \cap N(H)$ is finite and $N(H)/H$ is finite. Using the conjugacy theorem in H again, one again sees that the homomorphism

$$N(T) \cap N(H)/N(T) \cap H \rightarrow N(H)/H$$

induced by the preceding diagram is bijective on the points with values in k (in fact, it is an isomorphism), so, since the second is finite, so is the first; hence $N(T) \cap H$ has finite index in $N(T)$, and thus contains $C = N(T)^0$, so $H \supset C$. Thus (i), (i bis), (ii), (iii) are equivalent conditions.

Let us prove that (ii) $\Rightarrow$ (vii). One sees at once that conditions (ii) and (vii) are each invariant under an extension $k \rightarrow k'$ of the base field, with k' algebraically closed, which allows one to suppose that k has infinite transcendence degree over its prime subfield. Then it is well known (and one verifies immediately) that there exists an element a of $T(K)$ such that the subgroup of $T(K)$ which it generates is dense in T for the Zariski topology. One easily concludes that $(\mathfrak{g}/\mathfrak{h})^T = (\mathfrak{g}/\mathfrak{h})^{\text{ad}(a)}$, and since by hypothesis the left-hand side is zero, one concludes (vii).

Let us prove (vii) $\Rightarrow$ (vi). This implication is contained in the following result, which makes 2.1 more precise:

Corollary 2.2. *Let G be a smooth algebraic group over a field k , H a smooth algebraic subgroup, N its normalizer in G , $\varphi : G \times H \rightarrow G$ the morphism defined by $\varphi(g, h) = \text{ad}(g)h = ghg^{-1}$, $\psi : X = G \times^N H \rightarrow G$ the morphism deduced from φ by passage to the quotient (cf. no. 1), and $a \in H(k)$. The following conditions are equivalent:*

- (i) φ is smooth at (e, a) .
- (ii) ψ is étale at $(\bar{e}, a)$, and N is smooth over k .
- (iii) $(\mathfrak{g}/\mathfrak{h})^{\text{ad}(a)} = 0$ (where $\mathfrak{g}, \mathfrak{h}$ are the Lie algebras of G, H).

These conditions imply $H^0 = N^0$.

One knows that the smoothness of φ (which is a morphism of smooth k -preschemes) at a point rational over k is equivalent to the surjectivity of the tangent map at that point. Now an immediate calculation shows that this tangent map is written (by means of the usual identifications of the tangent spaces at points of G and H with the Lie algebra of G and H)

$$d\varphi(\xi, \eta) = (\text{id} - \text{ad}(a)) \cdot \xi + \eta,$$

considered as a map from $\mathfrak{g} \times \mathfrak{h}$ to $\mathfrak{g}$. Surjectivity is therefore equivalent to the surjectivity of $(\text{id} - \text{ad}(a))$ on $\mathfrak{g}/\mathfrak{h}$, i.e. to (iii). Now (iii) implies a fortiori $(\mathfrak{g}/\mathfrak{h})^H = 0$, i.e. (cf. 2.0, where the connectedness hypothesis at the beginning of the section is unnecessary) $H^0 = N^0$. One deduces that N is smooth, and $\dim H = \dim N$, whence $\dim X = \dim G$. Now, since $q : G \times H \rightarrow X$ is flat and $\psi = \varphi \circ q$ smooth at (e, a) , it follows that ψ is smooth at $q(e, a) = (\bar{e}, a)$, hence étale at this point for reasons of dimension. Thus one has proved (i) $\Leftrightarrow$ (iii) $\Rightarrow$ (ii); on the other hand (ii) $\Rightarrow$ (i), since the smoothness of N implies that of q . Q.E.D.

Finally, let us prove (vi) $\Rightarrow$ (i), which, together with the implications already established, will prove the theorem. Suppose first that G is affine. Let U be a nonempty open subset of G such that $x \in U(k)$ implies that x is contained in a conjugate of H . Let C be a Cartan subgroup of G . Using the implication (i) $\Rightarrow$ (iv) for C in place of H (this is Borel's "density theorem"), it follows that one can find a conjugate of C which meets U ; hence one may suppose that $U \cap C \neq \emptyset$, i.e. that there exists a nonempty open subset V of C such that for every $x \in V(k)$, x is contained in a conjugate of H . Write C as a product

$$C = T \cdot C_u$$

where T is the maximal torus of C (which is a maximal torus of G) and C_u the unipotent part of C , T being in the center of C (BIBLE 6 th. 2). One may again suppose that k has infinite transcendence degree over its prime subfield, which allows one to find an element t of $T(k)$ belonging to the projection of V onto T (which is a nonempty open subset of T), i.e. $t \cdot C_u \cap V \neq \emptyset$, and such that t "generates" T . Since every algebraic subgroup of G which contains a product $t \cdot u$ ($t \in T(k)$, $u \in C_u(k)$) contains both factors (BIBLE 4 th. 3), it follows, with the preceding choice of t , and taking $t \cdot u \in V(k)$, that there exists a conjugate of H which contains t , hence T . Thus one may already suppose that one has

$$T \subset H.$$

If W is the open subset of C_u which is the inverse image of V under $u \mapsto t \cdot u$, one therefore sees that for every element x of $(T \cdot W)(k)$, there exists a conjugate of H which contains T and x . As one has already noted, such a conjugate is of the form $\text{int}(g) \cdot H$, where $g \in N(T)(k)$. Consider then the morphism

$$f : N(T) \times H \rightarrow G$$

defined by $f(g, h) = \text{int}(g) \cdot h = ghg^{-1}$; then the image of f contains $T \cdot W$, so, since $N(T)$ is a finite union of translates $C \cdot g_i$ (where $g_i \in N(T)(k)$), as C has finite index in $N(T)$, it follows that there exists a dense open subset V' of $C = T \cdot C_u$ which is contained in the image of $(C \cdot g_i) \times H$ under f . Replacing H by $\text{int}(g_i) \cdot H$, one may suppose $g_i = e$, i.e. $f(C \times H) \supset V'$. Thus for every $u \in V'(k)$, there exist $v \in C(k)$ and $h \in H(k)$ such that

$$v^{-1}hv = u \quad \text{whence } vuv^{-1} \in H(k),$$

whence, setting

$$C' = C \cap H = \text{Centr}_H(T),$$

$vuv^{-1} \in C'$, whence $\text{int}(v) \cdot C' \rightarrow u$. This proves that the union of the conjugates of C' in C (by elements of $C(k)$) is dense, which implies (as one has seen for the pair (G, H) in place of (C, C')) that C' has finite index in its normalizer in C . By BIBLE 7, the lemma of no. 1, it follows that $C' = C$, hence $C = H$, which proves (vi) $\Rightarrow$ (i) when G is affine.

In the general case, we proceed by induction on $n = \dim G$, the assertion being trivial if $n = 0$. Let Z be the center of G , and distinguish two cases:

1°) $\dim Z \cap H > 0$; then, setting $G' = G/Z \cap H$, one has $\dim G' < n$; on the other hand, hypothesis (vi) on H implies the same condition for the image H' of H in G' , so H' contains a Cartan subgroup C' of G' , hence H contains the inverse image C of C' , which is a Cartan subgroup by XII 6.6 e).

2°) $\dim Z \cap H = 0$, so the canonical morphism $H \rightarrow G/Z$ is a finite morphism, and since G/Z is affine by XII 6.1, it follows that H is affine, hence every homomorphism from H to an abelian variety is zero (and even every morphism of preschemes from H to an abelian

variety is zero): this follows from the fact that a smooth connected affine algebraic group over an algebraically closed field is a rational variety, or merely that it is the union of its Borel subgroups (BIBLE 6 th. 5 b)); now it follows very easily from the structure theorems, BIBLE 6.2 and 6.3, that a smooth connected solvable affine group is a rational variety. Let us now use Chevalley's structure theorem for G , according to which G is an extension of an abelian variety A by a smooth affine group. Then the image of H in A is zero, H being affine; on the other hand it is identical to A , since the union of its conjugates in A must be dense. Thus $A = 0$, so G is affine, and one is reduced to the case already treated. This completes the proof of 2.1. $\square$

Corollary 2.3. *Suppose that the equivalent conditions of 2.1 hold.*

- a) *Let $k(X)$ (resp. $k(G)$) be the field of rational functions of X (resp. G); then $k(X)$ is a finite separable extension of $k(G)$. Denote its degree by d .*
- b) *Let T be a maximal torus of G contained in H (which exists by form 2.1 (ii)), and let C be the corresponding Cartan subgroup of G . Then $C \subset H$. On the other hand, $\text{Norm}_G(T)$ is a smooth subgroup of G and $\text{Norm}_G(T) \cap \text{Norm}_G(H) = \text{Norm}_{\text{Norm}_G(H)}(T)$ is a smooth subgroup of it of finite index equal to d (defined in a)). The number of conjugates of H containing a given maximal torus or a given Cartan subgroup is equal to d .*
- c) *Let U be the largest open subset of G such that $\psi : X \rightarrow G$ induces a morphism $\psi^{-1}(U) \rightarrow U$ which is finite and étale. Then U is a dense open subset, and for $g \in G(k)$, one has $g \in U(k)$ if and only if there exist exactly d conjugates of H containing g , or again if and only if there exist at least d distinct conjugates H_i of H containing g such that for every i , $(\mathfrak{g}/\mathfrak{h}_i)^{\text{ad}(g)} = 0$ (where $\mathfrak{h}_i = \text{Lie}(H_i)$).*

Assertion a) follows from the fact that ψ is generically étale (which was stated at the end of 2.1); this also implies that the open subset U introduced in c) is nonempty, i.e. dense, and the two stated characterizations of the elements of $U(k)$ (taking into account that ψ is separated, X integral and G integral and normal, SGA1 I 10.11, and the fact that $(\mathfrak{g}/\mathfrak{h}_i)^{\text{ad}(g)} = 0$ means that ψ is étale at the point x_i of $\psi^{-1}(g)$ corresponding to H_i). If H contains the maximal torus T of G , then the centralizers of T in H and G have the same dimension and are smooth and connected (XII 6.6 b)), hence are equal, which proves that $C \subset G$. Moreover, one knows that the normalizer of T in a smooth group which contains it is smooth (XI, 2.4 bis), so $\text{Norm}_G(T)$ and $\text{Norm}_{\text{Norm}_G(H)}(T)$ are smooth (N.B. it was noted that $N = \text{Norm}_G(H)$ is smooth, at the end of statement 2.1); moreover, $\text{Norm}_N(T)$ contains C , which has finite index in $N(T)$, hence it has finite index in $N(T)$. Using the conjugacy theorem for maximal tori of H , one sees that the index in question is equal to the number of conjugates of H which contain T , or, what amounts to the same thing, which contain C . Now, since the union of the conjugates of C in G is dense (by 2.1 (i) $\Rightarrow$ (vi) applied to C in place of H), and the open subset U defined in c) is clearly stable under inner automorphisms, one sees that $C \cap U \neq \emptyset$. Proceeding as in the proof of implication (vi) $\Rightarrow$ (ii) of 2.1, one concludes that (making, if necessary, a harmless change of base field) there exists a $g \in (C \cap U)(k)$ such that every conjugate of H which contains g contains T , and consequently also C . Thus the conjugates of H containing C are those containing g , and since $g \in U(k)$, their number is equal to d , which finishes establishing b). One will note that we have in fact established that the set of conjugates of H containing T is a homogeneous set under the group of rational points of

$$W_G(T) = \text{Norm}_G(T) / \text{Centr}_G(T),$$

which proves in particular that

$$d \leq \text{order of the Weyl group of } G.$$

Corollary 2.4. *With the notation of 2.1, the following conditions are equivalent:*

- (i) $\psi : X \rightarrow G$ is a birational morphism.
- (ii) *There is one and only one conjugate of H containing a given Cartan subgroup of G .*

- (iii) H contains a Cartan subgroup C of G , and $\text{Norm}_G(H) \supset \text{Norm}_G(C)$.
- (iv) There exists a nonempty open subset V of G such that $g \in V(k)$ implies that g is contained in exactly one conjugate of H .

This is clear by 2.1 and 2.3.

Corollary 2.5. *Suppose that the conditions of 2.4 hold, and let $g \in G(k)$. The following conditions are equivalent:*

- (i) $g \in U(k)$, where U is defined in 2.3 c), i.e. g is contained in one and only one conjugate of H .
- (ii) The set of conjugates of H containing g is finite and nonempty.
- (iii) The scheme $\psi^{-1}(g)$ “of conjugates of H containing g ” contains an isolated point.
- (iv) There exists a conjugate H' of H containing g , and one has $(\mathfrak{g}/\mathfrak{h}')^{\text{ad}(g)} = 0$, where $\mathfrak{h}' = \text{Lie}(H')$.

Finally, U is also the largest open subset of G such that ψ induces an isomorphism $\psi^{-1}(U) \xrightarrow{\sim} U$.

The equivalence of (i), (ii), (iii), as well as the last assertion, follows from the “Main Theorem”⁽³⁾ applied to the birational morphism $\psi : X \rightarrow G$, taking into account that G is normal. The equivalence of these conditions with (iv) follows at once from the last assertion of 2.1 characterizing the set of elements of X at which ψ is étale.

Theorem 2.6. *Let G be a smooth connected algebraic group over an algebraically closed field k , C a Cartan subgroup, associated to a maximal torus T , $N = \text{Norm}_G(C) = \text{Norm}_G(T)$ (cf. XII 8.4), and let $X = G \times^N C$, where N acts on the left factor G by right translations, and on the right factor C by inner automorphisms; let $\psi : X \rightarrow G$ be the canonical morphism.*

- a) *The morphism ψ is birational.*
- b) *Let U be the largest open subset of G such that ψ induces an isomorphism $\psi^{-1}(U) \xrightarrow{\sim} U$ (cf. 2.5). Let*

$$\rho = \rho_\nu(G) = \dim C$$

be the nilpotent rank of G . Then for every $g \in G(k)$, the multiplicity of the eigenvalue 1 in $\text{ad}(g)$ acting on $\mathfrak{g}$ is at least ρ , and for it to equal ρ , it is necessary and sufficient that one have $g \in U(k)$.

PROOF. Since condition 2.1 (i) is verified, one may apply 2.4 (iii) $\Rightarrow$ (i), which establishes (a). In BIBLE 7 (in the case where G is affine) the points of $U(k)$ are called the *regular points* of $G(k)$, and we shall follow this terminology by calling U the open subset of regular points of G . (N.B. The proof given in BIBLE of the fact that the set in question is itself open is incorrect, but we have obtained it in the present exposé under more general conditions.)

Let us prove b), and for this purpose introduce for every $g \in G(k)$ the characteristic polynomial

$$P(\text{ad}(g), t) = t^n + c_1(g)t^{n-1} + \cdots + c_n(g);$$

one sees at once (replacing k by an arbitrary algebra over k) that the $c_i(g)$ come from well-determined sections

$$c_i \in \Gamma(G, \mathcal{O}_G).$$

When $g \in G(k)$ is an element contained in a Cartan subgroup (for example, a regular element), which we may suppose to be C , then by 2.5 (iv) one sees that one has $(\mathfrak{g}/\mathfrak{c})^{\text{ad}(g)} = 0$ if and only if g is regular (where $\mathfrak{c}$ denotes the Lie algebra of C); on the other hand, since C is nilpotent one sees immediately that $\text{ad}_{\mathfrak{c}}(g)$ has only the eigenvalue 1, which proves that the multiplicity of the eigenvalue 1 in $\text{ad}_{\mathfrak{g}}(g)$ is $\geq \rho$, and equal to exactly $\dim C = \rho$ if and only if g is regular. In particular, the above polynomial is divisible by $(t - 1)^\rho$. Since the divisibility relation by $(t - 1)^\rho$ is expressed by linear relations (with integer coefficients) between the

⁽³⁾N.D.E. Of Zariski!

coefficients of the polynomial, and these relations hold for $g \in U(k)$, U being a dense open subset, it follows (G being reduced) that they hold for every g ; in fact one has a relation

$$(\dagger) \quad t^n + c_1 t^{n-1} + c_0 = (t-1)^\rho (t^{n-\rho} + b_1 t^{n-\rho-1} + \cdots + b_{n-\rho})$$

in the polynomial ring over $\Gamma(G, \mathcal{O}_G)$; in particular for every $g \in G(K)$, $\text{ad}(g)$ has the eigenvalue 1 with multiplicity at least ρ . Moreover, one has seen that one has equality if g is regular; let us prove the converse. For this, suppose first that G is affine, and write g as the product

$$g = g_s g_u$$

of its semisimple part and its unipotent part (BIBLE 4 no. 4); then

$$\text{ad}(g) = \text{ad}(g_s) \text{ad}(g_u)$$

is the analogous decomposition of $\text{ad}(g)$ (*loc. cit.* cor. to th. 3), and consequently $\text{ad}(g)$ and $\text{ad}(g_s)$ have the same eigenvalues (counted with their multiplicities); in particular the eigenvalue 1 occurs with the same multiplicity in $\text{ad}(g)$ and in $\text{ad}(g_s)$.

On the other hand, by BIBLE 7 th. 2 cor. 1, g is regular if and only if g_s is. Thus, to prove b), one may suppose $g = g_s$, i.e. g semisimple, hence contained in a maximal torus by BIBLE 6 th. 5 c), and a fortiori in a Cartan subgroup, a case which has already been treated. This proves b) in the case where G is affine. In the general case, let $Z = \text{Centr}(G)_{\text{red}}$; then by XII 6.6 e) the Cartan subgroups of G are the inverse images of those of $G' = G/Z$, so g is regular in G if and only if its image g' in G' is regular in G' . On the other hand, since Z is smooth, the Lie algebra $\mathfrak{g}'$ of G' is precisely $\mathfrak{g}/\mathfrak{z}$, where $\mathfrak{z} = \text{Lie}(Z)$, and $\text{ad}(g')$ is precisely $\text{ad}(g)_{\mathfrak{g}/\mathfrak{z}}$, so the multiplicity of the eigenvalue 1 in $\text{ad}(g)$ is equal to $d = \dim Z$ plus the multiplicity of the eigenvalue 1 in $\text{ad}(g')$, whence it follows at once that the first is equal to the nilpotent rank of G if and only if the second is equal to the nilpotent rank of G' . Thus one is reduced to the case of G' ; now, G' being affine by XII 6.1, this case has already been treated. This completes the proof of 2.6. $\square$

Corollary 2.7. *With the notation of the preceding proof⁽⁴⁾, let*

$$b = 1 + b_1 + \cdots + b_{n-\rho} \in \Gamma(G, \mathcal{O}_G).$$

Then the open subset of regular points of G is given by

$$U = G_b$$

(the set of points of G at which b is invertible); in particular U is an affine open subset if G is affine.

Corollary 2.8. *Let H be a smooth connected algebraic subgroup of G containing a Cartan subgroup of G .*

a) Let C be an algebraic subgroup of H . For C to be a Cartan subgroup of H , it is necessary and sufficient that it be a Cartan subgroup of G .

b) Let $g \in G(k)$, and let d be the integer introduced in 2.3. For g to be a regular point of G , it is necessary and sufficient that there exist exactly d conjugates H_i of H containing g , and that for each i , g be a regular element of H_i , or again that there be at most d conjugates of H containing g , and that g be regular in one of them. If this is so, and if C is the unique Cartan subgroup of G containing g , then the conjugates of H containing g are the conjugates of H containing C .

c) Let $g \in H(k)$; for g to be regular in G , it is necessary and sufficient that it be regular in H , and that one have

$$(\mathfrak{g}/\mathfrak{h})^{\text{ad}(g)} = 0.$$

⁽⁴⁾**N.D.E.** Cf. ($\dagger$) above.

PROOF. Let us prove a). Under either hypothesis on C , the unique maximal torus T of C is a maximal torus of H and of G (H having the same reductive rank as G), so, since $\text{Centr}_H(T) \subset \text{Centr}_G(T)$ are smooth connected groups of the same dimension, they are equal; hence it amounts to the same thing to say that C is equal to one or the other of these two groups, which proves a).

Let us prove b). Suppose first that g is regular for G , and let C be the unique Cartan subgroup of G containing g ; then by 2.3 b) there exist exactly d conjugates H_i of H containing C . Since $(\mathfrak{g}/\mathfrak{c})^{\text{ad}(g)} = 0$, i.e. $\text{ad}(g)$ does not have the eigenvalue $+1$ on $\mathfrak{g}/\mathfrak{c}$, one has a fortiori $(\mathfrak{g}/\mathfrak{h}_i)^{\text{ad}(g)} = 0$, so by 2.3 c) there are exactly d conjugates of H containing g , namely the H_i . For such an H_i , a Cartan subgroup of H_i containing g is a Cartan subgroup of G containing g , by a), and is therefore equal to C , which proves that g is regular in H_i . Conversely, suppose that there exist at most d conjugates H_i of H containing g , and that g is regular in one of them, which may be supposed equal to H . Let us prove that g is regular in G . Since g is regular in H , it is contained in a unique Cartan subgroup C of H ; by a), this is a Cartan subgroup of G . Let C' be a Cartan subgroup of G containing g ; let us prove $C' = C$ (which will prove that g is regular in C). Indeed, by 2.3 b), there exist exactly d conjugates of H containing C' , and since these contain g , they are necessarily the H_i , so the H_i , and in particular H , contain C' . Thus C, C' are two Cartan subgroups of H (by a)) which contain the same regular element g of H , and are therefore equal. Q.E.D.

Let us prove c): denoting by $\nu(u)$ the nullity⁽⁵⁾ of $\text{id} - u$, for an endomorphism of a finite-dimensional vector space, one will have

$$\nu(\text{ad}(g)_{\mathfrak{g}}) = \nu(\text{ad}(g)_{\mathfrak{h}}) + \nu(\text{ad}(g)_{\mathfrak{g}/\mathfrak{h}}),$$

now the two terms on the right-hand side are respectively $\geq$ the nilpotent rank of H (equal to the nilpotent rank ρ of G , by a)) and ≥ 0 , so one has $\nu(\text{ad}(g)_{\mathfrak{g}}) = \rho$ if and only if $\nu(\text{ad}(g)_{\mathfrak{h}}) = \rho$ and $\nu(\text{ad}(g)_{\mathfrak{g}/\mathfrak{h}}) = 0$, i.e. g is regular in G if and only if g is regular in H and $\text{ad}(g)_{\mathfrak{g}/\mathfrak{h}}$ has no nonzero invariants. □

Remarks 2.9. In statement 2.1, one cannot weaken condition (iii) by supposing merely that H contains a maximal torus and has finite index in its normalizer, even if one requires that this normalizer moreover be smooth, i.e. that one have $H = N^0$, and even when G is affine and solvable. An example is provided by the group G of matrices of the form

$$g = \begin{pmatrix} t & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix},$$

and the subgroup H of matrices of the preceding form, with $b = c = 0$ (N.B. the Cartan subgroup of G here consists of the matrices g with $a = c = 0$).

Remarks 2.10. Let G be a smooth algebraic group over k , H a smooth algebraic subgroup, but no longer suppose that H and G are connected. Suppose that H^0 contains a Cartan subgroup of G^0 . Then $(\mathfrak{g}/\mathfrak{h})^H \subset (\mathfrak{g}/\mathfrak{h})^{H^0} = 0$, so $H^0 = N^0$ (N is the normalizer of H); in particular N is smooth. However, one easily constructs examples, with G connected, where H has a connected component H_i such that for no $h \in H_i(k)$ does one have $(\mathfrak{g}/\mathfrak{h})^{\text{ad}(h)} = 0$, i.e. the morphism $(g, h) \mapsto \text{ad}(g) \cdot h$ from $G \times H_i$ to G is étale (or even quasi-finite) at no point (take, for example, for H the normalizer of the maximal torus in $\text{SL}(2)_k$). Likewise, even if the image H' of H in the finite group $G' = G/G^0$ is equal to G' , it is not necessarily true that the union of the conjugates of H in G is dense (take for G the semidirect product of $\mathbf{Z}/2\mathbf{Z}$ with $G^0 = \text{SL}(2)_k$, on which it acts by “symmetry”, and for H the semidirect product $\mathbf{Z}/2\mathbf{Z} \cdot T$, where T is a maximal torus of G^0). On the other hand, if one does not assume a priori that H^0 contains a Cartan subgroup of G^0 , but that the union of the conjugates of H in G is dense, then H^0 necessarily contains a Cartan subgroup of G^0 : to verify this, one may

⁽⁵⁾**N.D.E.** That is, the dimension of its nilspace.

clearly suppose G connected, and it suffices to repeat the proof of 2.1 (vi) $\Rightarrow$ (i), which is valid without supposing H connected.

3. Case of an arbitrary base prescheme

Suppose first that one is over a base field k , not necessarily algebraically closed. Since conditions 2.1 (i bis), (iv bis), (v), (vi) are invariant under extension of the base field, one sees by passing to the algebraic closure $\bar{k}$ of k that they are equivalent to one another, and equivalent to the fact that $H_{\bar{k}}$ contains a Cartan subgroup of $G_{\bar{k}}$. When this condition is satisfied, then (with the notation of 2.3) it will still be true that $k(X)$ is a finite separable extension of $k(G)$, of degree d independent of any extension of the base. If U is the largest open subset of G such that ψ induces a morphism $\psi^{-1}(U) \rightarrow U$ which is finite and étale, then the formation of U commutes with extension of the base field. If ψ is birational, then U is also the largest open subset of G such that ψ induces an isomorphism $\psi^{-1}(U) \xrightarrow{\sim} U$, and if then $g \in U(k)$, there exists one and only one subgroup H' of G , conjugate to H over the algebraic closure $\bar{k}$ of k , such that $H' \ni g$.

A point $g \in G(k)$ is said to be *regular* if it is regular as an element of $G(\bar{k}) = G_{\bar{k}}(\bar{k})$. More generally, the construction of 2.7 gives one an open subset of G , whose formation commutes with every extension of the base field, called the *open subset of regular points* of G , which is also characterized by the fact that for every algebraically closed extension K of k and every point $g \in G(K)$, g is a regular point of G_K if and only if $g \in U(K)$. If $g \in U(k)^{(6)}$, one sees that g is contained in one and only one Cartan subgroup of G , as we shall show under more general conditions below.

Let G be a smooth separated group prescheme of finite type with connected fibers over the prescheme S ; consider the functor $\mathcal{C} : ((\text{Sch}))_S^\circ \rightarrow (\text{Ens})$ defined by

$$\mathcal{C}(S') = \text{set of Cartan subgroups (XII 3.1) of } G_{S'}.$$

Suppose that this functor is representable by a prescheme smooth over S ; we give in XV a necessary and sufficient condition for this to be so, but we already know that this hypothesis is satisfied if G is affine over S of locally constant reductive rank (XII 3.3), or more generally if G admits a maximal torus locally for the fpqc topology (XII 7.1 a)), for example if S is the spectrum of a field. Let X be the Cartan subgroup of the $\mathcal{C}$ -group prescheme $G_{\mathcal{C}}$, the “universal Cartan subgroup” of G . As a prescheme over S , X therefore represents the functor

$$X(S') = \{(C, g) : C \text{ a Cartan subgroup of } G_{S'} \text{ and } g \text{ a section of } C \text{ over } S'\}.$$

Consider the canonical projection morphism $(C, g) \mapsto g$

$$\psi : X \rightarrow G.$$

One then has the

Theorem 3.1. *Under the preceding conditions on G , and with the preceding notation, let U be the set of $g \in G$ such that g is a regular element of its fiber G_s . Then U is open, and it is also the largest open subset U of G such that ψ induces an isomorphism*

$$\psi^{-1}(U) \xrightarrow{\sim} U.$$

PROOF. Let us first prove that U is open. From the hypothesis of representability of $\mathcal{C}$ as a prescheme smooth over S , since its structural morphism is clearly surjective, one concludes at once that G admits a Cartan subgroup locally for the étale topology, and that the nilpotent rank of the fibers of G is locally constant. The same holds for the dimension of the fibers of

⁽⁶⁾N.D.E. Correction of $G(k)$ to $U(k)$.

G , and, localizing on S , one may suppose that both are constant, say ρ and n . Consider then the Killing polynomial

$$P_G(t) = t^n + c_1 t^{n-1} + \cdots + c_n \in A[t], \quad \text{where } A = \Gamma(G, \mathcal{O}_G).$$

The restriction of this polynomial to the fibers G_s of G , and in particular to the fibers at the maximal points of S , is divisible by $(t-1)^\rho$, which is expressed by the fact that certain linear combinations with integer coefficients of the c_i are zero on the fibers G_s . When S is reduced (which one may suppose in order to establish that U is open), it follows that they are themselves zero, hence that the Killing polynomial itself is divisible by $(t-1)^\rho$, say

$$P_G(t) = (t-1)^\rho (t^{n-\rho} + b_1 t^{n-\rho-1} + \cdots + b_{n-\rho}).$$

Let b be the sum of the coefficients $b_0 = 1, b_1, \dots, b_{n-\rho}$ of the second factor; then by 2.7 applied to the fibers of G , one sees that

$$U = G_b,$$

which indeed proves that U is open.

To prove that $\psi^{-1}(U) \rightarrow U$ is an isomorphism, one is reduced by SGA1 I 5.7 to verifying it fiber by fiber, which reduces one to the case of a base field, which may be supposed algebraically closed. Then there exists a Cartan subgroup C of G , and if N is its normalizer, C is identified by the conjugacy theorem XII 7.1 a) and b) with G/N , and the morphism $\psi : X \rightarrow G$ considered here is precisely that defined in no. 2. One then concludes by 2.6 b). The same argument also shows that U is the largest open subset of G such that ψ induces an isomorphism $\psi^{-1}(U) \rightarrow U$. $\square$

Corollary 3.2. *Under the conditions of 3.1, let g be a regular section of G , i.e. such that for every $s \in S$, $g(s)$ is a regular point of G_s . Then there exists one and only one Cartan subgroup C of G such that g is a section of C .*

Indeed, the hypothesis on g means that g is a section of U , and the conclusion that there exists a unique section of X lifting it, which is merely another way of expressing that $\psi^{-1}(U) \rightarrow U$ is an isomorphism.

Note now that the open subset $\psi^{-1}(U)$ of the Cartan subgroup X of G_C is precisely the open subset of X consisting of the points of X which are regular in G_C (by which one means: regular in their fiber). One thus obtains a natural “fibration” of the dense open subset U of regular points of G over the prescheme $\mathcal{C}$, the fibers being dense open subsets of Cartan subgroups of the fibers of G_C (namely the open subsets of regular points in G). One finds, for example, the following result (which will be made considerably more precise in the next exposé):

Corollary 3.3. *Let G be a smooth connected algebraic group over the field k , $\mathcal{T}$ the scheme of maximal tori of G ($\simeq$ the scheme of Cartan subgroups of G). Then the function field $k(G)$ of G is isomorphic to the function field of a smooth connected affine nilpotent algebraic group C over the function field $k(\mathcal{T})$ of $\mathcal{T}$, namely $C =$ “the generic Cartan subgroup of G ”. If G is affine of unipotent rank zero, i.e. if the Cartan subgroups of G_K are tori, then $k(G)$ is a unirational extension of $k(\mathcal{T})$.*

Of course, by the generic Cartan subgroup of G , one means (by abuse of language) the Cartan subgroup of $G_{k(\mathcal{T})}$ which is the generic fiber of X over $\mathcal{T}$. It only remains to prove the last assertion of 3.3, which is contained in the following well-known result (due to Chevalley):

Lemma 3.4. *Let k be a field, T a torus over k , and $k(T)$ the field of rational functions on T ; then $k(T)$ is a unirational extension of k , i.e. is contained in a purely transcendental extension of k .*

Indeed, let k' be a finite separable extension of k which splits⁽⁷⁾ T (X 1.4); then $T \otimes_k k'$ is a rational variety, i.e. admits a dense open subset isomorphic to a dense open subset of affine space $\mathbb{A}_{k'}^n$, hence $T' = \prod_{\text{Spec}(k')/\text{Spec}(k)} T_{k'}/\text{Spec}(k')$ is a rational variety (for it admits a dense open subset isomorphic to a dense open subset of $\prod_{\text{Spec}(k')/\text{Spec}(k)} \mathbb{A}_{k'}^n$, which is isomorphic to affine space of dimension mn over k , where $m = [k' : k]$). Consider the norm homomorphism from T' to T (defined as soon as T is a commutative group scheme over k); the composite $T \rightarrow T' \rightarrow T$ is the m th power on T , hence dominant, so $T' \rightarrow T$ is dominant, which proves that T is unirational.

Let us return to the conditions of 3.1, but supposing even that G admits a maximal torus locally for the fpqc topology (XII 7.1). Let Y be the maximal torus of the Cartan subgroup X of G , so that the morphism $\psi : X \rightarrow G$ induces a morphism $Y \rightarrow G$ whose image consists, as a set, of the semisimple elements of the fibers of G (XII 8). Finally, it follows from 3.1 that the restriction of ψ to the open subset Y^{reg} of regular points of Y induces a closed immersion

$$Y^{\text{reg}} \rightarrow U = G^{\text{reg}}.$$

Making explicit the meaning of $Z = Y^{\text{reg}}$ as a functor over S , one finds:

Corollary 3.5. *Let G be a smooth separated S -group prescheme of finite type with connected fibers over the prescheme S , admitting a maximal torus locally for the fpqc topology. Let $Z : ((\text{Sch}))_S^\circ \rightarrow (\text{Ens})$ be the functor defined by*

$$Z(S') = \{\text{regular sections of } G_{S'} \text{ over } S' \text{ which are contained in a maximal torus of } G_{S'}\}.$$

Then Z is representable by a closed subprescheme, smooth over S , of the open subset $U = G^{\text{reg}}$ of G introduced in 3.1.

Finally, note the following result, which makes the density theorem 2.1 (i) $\Rightarrow$ (vi) more precise:

Corollary 3.6. *Under the conditions of 3.5, let C be a Cartan subgroup of G , and consider the morphism*

$$\varphi : Z \times C \rightarrow G$$

defined by $\varphi(g, h) = \text{ad}(g)h = ghg^{-1}$. Then φ is dominant.

It clearly suffices to prove this fiber by fiber, which reduces one to the case where S is the spectrum of an algebraically closed field. Let T be the maximal torus of C , t_0 an element of $T(k)$ regular in G , and c_0 an element of $C(k)$ regular in G ; consider $\varphi^{-1}(\varphi(t_0, c_0))$, whose points rational over k are the pairs (t, c) , with $t \in Z(k)$, $c \in C(k)$, such that

$$\text{ad}(t)c = \text{ad}(t_0)c_0 \quad \text{i.e.} \quad c = \text{ad}(t^{-1}t_0)c_0,$$

which are therefore in one-to-one correspondence with the $t \in Z(k)$ such that $\text{ad}(t^{-1}t_0)c_0 \in C$, or, what amounts to the same thing, c_0 being regular, such that $t^{-1}t_0 \in N$ (the normalizer of C), i.e. $t \in N$. One obtains an open and closed part of this fiber by restricting to the $t \in Z(k)$ such that $t \in C(k)$. Thus one has found a connected component of $\varphi^{-1}(\varphi(t_0, c_0))$ isomorphic to T (N.B. that the preceding set-theoretic argument does indeed give an isomorphism of schemes is seen by replacing points with values in k by points with values in an arbitrary k -prescheme), so the generic fiber of φ has dimension $\leq \dim T$, hence $\text{Im}(\varphi)$ has dimension $\geq \dim Z \times C - \dim T = \dim Z + \dim C - \dim T$; now one has $\dim Z = \dim Y = \dim \mathcal{C} + \dim T = \dim G - \dim C + \dim T$, whence finally $\dim \text{Im}(\varphi) \geq \dim G$, so φ is dominant. Q.E.D.

Remarks 3.7. One will note that the argument shows in addition that the connected component at (t_0, c_0) of the fiber of $\varphi(t_0, c_0)$ is isomorphic to T , in particular is smooth over k , and has the same dimension as the generic fiber, which implies that φ is in fact smooth at

⁽⁷⁾**N.D.E.** Splits (?), or: “over which T is split”.

(t_0, c_0) (which one should also be able to verify by calculating the tangent map). It follows that under the conditions of 3.6 the induced morphism $Z \times_S C^{\text{reg}} \rightarrow G^{\text{reg}}$ (where one has set $C^{\text{reg}} = C \cap G^{\text{reg}}$) is a smooth morphism. One sees likewise that the analogous morphism $Z \times T^{\text{reg}} \rightarrow Z$ (where T is a maximal torus of G) is smooth; more generally, for every smooth connected invariant algebraic subgroup H of C containing a regular element c_0 of $G(k)$, the image of $Z \times H \rightarrow G$ is dense in that of $G \times H \rightarrow G$.

4. Lie algebras over a field: rank, regular elements, Cartan subalgebras

In the remainder of this exposé, we take up again the theory developed by Chevalley in his book “Théorie des Groupes de Lie III” (Act. Sc. Ind. 1226, Paris 1955), the technique of schemes allowing us to eliminate the characteristic-zero hypothesis. We begin by recalling in this section certain well-known notions and results.

Let $\mathfrak{g}$ be a Lie algebra over a ring k . For every $a \in \mathfrak{g}$, one denotes by $\text{ad}(a)$ the endomorphism

$$\text{ad}(a) \cdot x = [a, x]$$

of $\mathfrak{g}$, which is a derivation of the Lie algebra $\mathfrak{g}$. Now, for every derivation D of the Lie algebra $\mathfrak{g}$, the nilspace of D , i.e. the union of the kernels of the iterates of D , is a Lie subalgebra of $\mathfrak{g}$, as one sees from Leibniz’s formula

$$D^m([x, y]) = \sum_{0 \leq p \leq m} \binom{m}{p} [D^p x, D^{m-p} y].$$

We shall set

$$\text{Nil}(a, \mathfrak{g}) = \text{nilspace of } \text{ad}(a) = \bigcup_{m \geq 0} \text{Ker } \text{ad}(a)^m;$$

when no confusion is to be feared, we shall denote it simply by $\text{Nil}(a)$, and call it the nilspace of a (in $\mathfrak{g}$).

Proposition 4.1. *For every $a \in \mathfrak{g}$, its nilspace $\text{Nil}(a, \mathfrak{g})$ is a Lie subalgebra of $\mathfrak{g}$, equal to its own normalizer.*

It remains to prove that it is its own normalizer, i.e. that every element of $\mathfrak{g}/\text{Nil}$ annihilated by the adjoint representation of Nil in $\mathfrak{g}/\text{Nil}$ is zero, which is trivial (for every element in this quotient annihilated by $\text{Ad}(a)$ is zero).

In the rest of this section, we suppose that k is a field, and $\mathfrak{g}$ finite-dimensional over k . We shall denote by $W(\mathfrak{g})$ the scheme over k defined by $\mathfrak{g}$, whose points in the k -algebra A are the elements of $\mathfrak{g} \otimes_k A$. If $a \in \mathfrak{g}$, the characteristic polynomial of $\text{ad}(a)$ is also called the characteristic polynomial or *Killing polynomial* of a in $\mathfrak{g}$, say

$$P_{\mathfrak{g}}(a, t) = t^n + c_1(a)t^{n-1} + \cdots + c_n(a),$$

where $n = \text{rank}_k \mathfrak{g}$, the $c_i(a) \in k$. Taking this polynomial also for $a \in \mathfrak{g} \otimes_k A^{(8)}$, where A is an arbitrary k -algebra, one sees that the $c_i(a)$ come from well-determined sections c_i of the structure sheaf of $W(\mathfrak{g})$, i.e. elements of the symmetric algebra $A = \text{Sym}_k(\mathfrak{g}^\vee)$, where $\mathfrak{g}^\vee$ is the dual of the k -module $\mathfrak{g}$. (When k is an infinite field, the c_i are determined by knowledge of the corresponding polynomial functions $\mathfrak{g} \rightarrow k$, but this is no longer so if k is a finite field.) Let r be the largest integer such that the Killing polynomial

$$P_{\mathfrak{g}}(t) = t^n + c_1 t^{n-1} + \cdots + c_n \in A[t]$$

is divisible by t^r , i.e. one has:

$$P_{\mathfrak{g}}(t) = t^n + c_1 t^{n-1} + \cdots + c_{n-r} t^r, \quad c_{n-r} \neq 0.$$

The integer r is called the *nilpotent rank* of the Lie algebra $\mathfrak{g}$. It is invariant under extension of the base field.

⁽⁸⁾**N.D.E.** $W \otimes_k A$ has been corrected to $\mathfrak{g} \otimes_k A$.

Proposition 4.2. *Let r be the nilpotent rank of $\mathfrak{g}$, and let $a \in \mathfrak{g}$. Then one has*

$$\text{rank}_k \text{Nil}(a, \mathfrak{g}) \geq r,$$

and one has equality if and only if one has

$$c_{n-r}(a) \neq 0.$$

In this case, $\text{Nil}(a, \mathfrak{g})$ is a nilpotent Lie algebra (and we shall see in 5.7 b) the converse, when $\mathfrak{g}$ is the Lie algebra of a smooth algebraic group G over k).

The first assertion is trivial, for by definition one has $\text{rank}_k \text{Nil}(a, \mathfrak{g}) =$ the multiplicity of the zero root in $P_{\mathfrak{g}}(a, t)$. Let us prove that if $c_{n-r}(a) \neq 0$, then $\text{Nil}(a)$ is nilpotent, which also means that for every $x \in \text{Nil}(a)$, $\text{ad}_{\text{Nil}(a)}(x)$ is a nilpotent endomorphism. One may suppose k algebraically closed; then, since $\text{ad}(a)_{\mathfrak{g}/\text{Nil}(a)}$ is injective, there exists a nonempty open subset U of $W(\text{Nil}(a))$ such that for every $x \in U(k)$, $\text{ad}(x)_{\mathfrak{g}/\text{Nil}(a)}$ is injective, hence $\text{Nil}(x) \subset \text{Nil}(a)$; one may moreover suppose U contained in the open subset of points where c_{n-r} does not vanish (since this open subset is nonempty by $c_{n-r}(a) \neq 0$), and then $\text{Nil}(x)$ having the same dimension as $\text{Nil}(a)$, one will have $\text{Nil}(x) = \text{Nil}(a)$. Consequently, for every $x \in U(k)$, $\text{ad}(x)_{\text{Nil}(a)}$ is nilpotent, and by the principle of extension of algebraic identities, this will remain true for every $x \in \text{Nil}(a)$, so $\text{Nil}(a)$ is nilpotent.

One says that the element a of $\mathfrak{g}$ is *regular* if $c_{n-r}(a) \neq 0$, i.e. if $\text{rank}_k \text{Nil}(a, \mathfrak{g}) = r$. When k is infinite, this therefore also means that $\text{rank}_k \text{Nil}(a, \mathfrak{g})$ is as small as possible (for a varying in $\mathfrak{g}$). In all cases, the notion of regular element of $\mathfrak{g}$ is invariant under extension of the base field, and the set of points of $W(\mathfrak{g})$ which are regular (i.e. which come from regular points of $W(\mathfrak{g})$ with values in a suitable extension of k) is open, for it is identical to $W(\mathfrak{g})_{c_{n-r}}$ (the set of points where c_{n-r} is invertible).

Corollary 4.3. *Let a be a regular element of $\mathfrak{g}$, and $\mathfrak{h}$ a Lie subalgebra of $\mathfrak{g}$ containing a . Then $\mathfrak{h}$ is nilpotent if and only if $\mathfrak{h} \subset \text{Nil}(a, \mathfrak{g})$; in particular, $\text{Nil}(a, \mathfrak{g})$ is a maximal nilpotent subalgebra of $\mathfrak{g}$.*

Since $\text{Nil}(a)$ is nilpotent, the relation $\mathfrak{h} \subset \text{Nil}(a)$ indeed implies that $\mathfrak{h}$ is nilpotent, and conversely, if $\mathfrak{h}$ is nilpotent, it is contained in the nilspace of its element a , i.e. $\mathfrak{h} \subset \text{Nil}(a)$.

Proposition 4.4. *Suppose k infinite. Let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{g}$. Consider the following conditions:*

- (i) $\mathfrak{d}$ is maximal nilpotent and contains a regular element of $\mathfrak{g}$.
- (i bis) $\mathfrak{d}$ is of the form $\text{Nil}(a, \mathfrak{g})$, where a is a regular element of $\mathfrak{g}$.
- (ii) $\mathfrak{d}$ is nilpotent and of the form $\text{Nil}(a, \mathfrak{g})$, where $a \in \mathfrak{g}$.
- (ii bis) $\mathfrak{d}$ is nilpotent, and there exists $a \in \mathfrak{d}$ such that $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective.
- (iii) $\mathfrak{d}$ is nilpotent and identical to its own normalizer.

One has the implications:

$$(i) \Leftrightarrow (i \text{ bis}) \Rightarrow (ii) \Leftrightarrow (ii \text{ bis}) \Leftrightarrow (iii)$$

(and we shall see in 5.7 a) that if $\mathfrak{g}$ is the Lie algebra of a smooth algebraic group, then all the preceding conditions are equivalent).

The equivalence of (i) and (i bis) is trivial by 4.3, and these conditions trivially imply (ii). The equivalence of (ii) and (ii bis) is also trivial, as is (ii bis) $\Rightarrow$ (iii) (cf. 4.1). It remains to prove implication (iii) $\Rightarrow$ (ii bis), the only one, moreover, which uses the fact that k is infinite, and which follows at once from the

Lemma 4.5. *Let $\mathfrak{d}$ be a nilpotent Lie algebra over an infinite field k , acting on a finite-dimensional vector space V . Suppose that for every $x \in \mathfrak{d}$, the endomorphism $u(x)$ is not injective. Then there exists a nonzero element v of V annihilated by $\mathfrak{d}$.*

One may suppose k algebraically closed and $\mathfrak{d}$ finite-dimensional. One then knows that V is a direct sum of finitely many nonzero stable vector subspaces V_i ($1 \leq i \leq n$), such that for every i , and every $x \in \mathfrak{d}$, $u(x)|_{V_i}$ has a single eigenvalue $\lambda_i(x)$ (cf. Bourbaki, *Groupes et Algèbres de Lie*, Chap. I, §4, Exercise 22). Let $c_i(x)$ be the constant term of the characteristic polynomial of $u(x)|_{V_i}$, so that $\lambda_i(x) = 0$ if and only if $c_i(x) = 0$. Then c_i is a polynomial function on $\mathfrak{d}$, and the hypothesis means that $\mathfrak{d}$ is the union of the zero sets of the c_i . Thus one of the c_i is zero, which reduces one (replacing V by V_i) to the case where V is such that the $u(x)$ ($x \in \mathfrak{d}$) are nilpotent. But then Engel's theorem (Bourbaki *loc. cit.* th. 1) implies that there exists a nonzero v in V annihilated by $\mathfrak{d}$. Q.E.D.

One easily sees that (k still being an infinite field) conditions (i), (i bis) of 4.4 are invariant under every extension of the base field. If they hold, one will say that $\mathfrak{d}$ is a *Cartan subalgebra* of $\mathfrak{g}$; in the general case (k not necessarily infinite) one will say that $\mathfrak{d}$ is a Cartan subalgebra of $\mathfrak{g}$ if it becomes a Cartan subalgebra for one (and consequently every) extension of the base field $k \rightarrow k'$, with k' infinite. This therefore implies that $\mathfrak{d}$ is nilpotent and equal to its own normalizer.

Proposition 4.6. *a) Let a be an element of $\mathfrak{g}$. If a is regular, it is contained in one and only one Cartan subalgebra of $\mathfrak{g}$ (and we shall see in 6.1 d) the converse when k is algebraically closed and $\mathfrak{g}$ the Lie algebra of a smooth algebraic group).*

b) Let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$, and a an element of $\mathfrak{d}$; then a is regular in $\mathfrak{g}$ if and only if $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective.

Indeed, for a) one notes that if a is regular, then $\text{Nil}(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$ (for this is true over an infinite extension k' of k), and it then follows at once from 4.3 that every Cartan subalgebra of $\mathfrak{g}$ containing a is identical to the preceding one. For b) one notes that the nullity⁽⁹⁾ of $\text{ad}(a)_{\mathfrak{g}}$ is equal to the sum of the nullities of $\text{ad}(a)_{\mathfrak{d}}$ and of $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$, and since the first is r , the sum is equal to r if and only if $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective. Q.E.D.

Corollary 4.7. *Let a be a regular element of $\mathfrak{g}$, $\mathfrak{d}$ a Cartan subalgebra of $\mathfrak{g}$ containing a , A an algebra over k , $\mathfrak{g}_A$ and $\mathfrak{d}_A \subset \mathfrak{g}_A$ the A -Lie algebras obtained from $\mathfrak{g}, \mathfrak{d}$ by base change, and a_A the image of a in $\mathfrak{g}_A$. Let u be an automorphism of $\mathfrak{g}_A$. For $u(\mathfrak{d}_A) = \mathfrak{d}_A$ to hold, it is necessary and sufficient that one have $u(a_A) \in \mathfrak{d}_A$.*

The condition is trivially necessary; let us prove that it is also sufficient. If it holds, then $\mathfrak{d}' = u(\mathfrak{d}_A)$ is a Lie subalgebra containing a_A , and every element b of it is such that $\text{ad}(b)_{\mathfrak{d}}$ is nilpotent (for $\mathfrak{d}'$ is isomorphic to $\mathfrak{d}_A$, which has this property, as follows at once from the definition of “nilpotent” in Bourbaki, *Groupes et Algèbres de Lie*, Chap. I, §4, def. 1). Taking $b = a_A$, one sees that the nilspace $\text{Nil}(b, \mathfrak{g}_A)$ contains $\mathfrak{d}'$; on the other hand, it is equal to $\mathfrak{d}_A$, and since $\mathfrak{d}'$ is locally a direct summand in the module $\mathfrak{g}_A$ ($\mathfrak{d}$ being so), hence in $\mathfrak{d}_A$, and it is a projective module of the same rank r as the latter, one concludes that it is equal to it. Q.E.D.

Proposition 4.8. *Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$.*

a) The following conditions are equivalent if k is infinite:

- (i) $\mathfrak{h}$ contains a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$.
- (ii) $\mathfrak{h}$ contains a regular element a of $\mathfrak{g}$, and an element b such that $\text{ad}(b)_{\mathfrak{g}/\mathfrak{h}}$ is injective.
- (iii) $\mathfrak{h}$ has the same nilpotent rank as $\mathfrak{g}$, and contains a regular element of $\mathfrak{g}$.

These conditions are invariant under extension of the base field k .

b) Suppose that these conditions hold over a suitable infinite extension k' of k . Let $a \in \mathfrak{h}$; then a is regular in $\mathfrak{g}$ if and only if it is regular in $\mathfrak{h}$ and $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective, i.e. if and only if $\text{Nil}(a, \mathfrak{h}) = \text{Nil}(a, \mathfrak{g})$ and this is a Cartan subalgebra of $\mathfrak{h}$.

c) Under the conditions of b), let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{h}$; for it to be a Cartan subalgebra of $\mathfrak{h}$, it suffices that it be a Cartan subalgebra of $\mathfrak{g}$ (and we shall see in 5.8 that the condition

⁽⁹⁾**N.D.E.** That is, the dimension of the nilspace.

is also necessary if $\mathfrak{g}$ is the Lie algebra of a smooth algebraic group G , and $\mathfrak{h}$ the Lie algebra of a smooth algebraic subgroup H of G).

One sees immediately that conditions (ii) and (iii) of a) are invariant under extension of the base field k (supposed infinite), and that in statements b) and c), one may suppose k infinite, which we shall do. If $\mathfrak{h}$ contains the Cartan subalgebra $\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$, then a is a regular element of $\mathfrak{g}$ such that $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective, hence (i) $\Rightarrow$ (ii). Conversely, if (ii) holds, then for a “sufficiently general” element a of $\mathfrak{h}$, a simultaneously satisfies the two conditions considered in (ii), hence $\text{Nil}(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$ and is contained in $\mathfrak{h}$, so one has (i). Thus (i) and (ii) are equivalent. Suppose they hold, and let a be a variable element of $\mathfrak{h}$; then

$$(x) \quad \begin{aligned} \text{rank}_k \text{Nil}(a, \mathfrak{g}) &= \text{rank}_k \text{Nil}(a, \mathfrak{h}) \\ &\quad + \text{rank}_k \text{Nil}(\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}), \end{aligned}$$

on the other hand, the two terms on the right-hand side are respectively $\geq r' =$ the nilpotent rank of $\mathfrak{h}$, and ≥ 0 , the equalities moreover being attained⁽¹⁰⁾ for a “sufficiently general” element of $\mathfrak{h}$. Moreover, one also has $\text{rank}_k \text{Nil}(a, \mathfrak{g}) \geq r =$ the nilpotent rank of $\mathfrak{g}$, equality being attained for a “sufficiently general” element of $\mathfrak{h}$, and being attained if and only if a is regular in $\mathfrak{g}$. One concludes from this that one has $r = r'$, and that a is regular if and only if the two terms on the right-hand side of (x) are equal respectively to r' and to 0, i.e. if and only if a is regular in $\mathfrak{h}$ and $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective, which proves b); and c) follows trivially by taking an element a in $\mathfrak{d}$ regular in $\mathfrak{g}$, so that $\text{Nil}(a, \mathfrak{g}) = \mathfrak{d}$. Moreover, the preceding result shows that (i) $\Rightarrow$ (iii); finally (iii) $\Rightarrow$ (i), for under (iii), a sufficiently general element a of $\mathfrak{h}$ is regular in $\mathfrak{h}$ and in $\mathfrak{g}$, so $\text{Nil}(a, \mathfrak{h}) \subset \text{Nil}(a, \mathfrak{g})$ are respectively Cartan subalgebras of $\mathfrak{h}$ and of $\mathfrak{g}$, and since they have the same rank over k , they are identical, which proves (i). This completes the proof of 4.8.

5. Case of the Lie algebra of a smooth algebraic group: density theorem

Let G be a smooth algebraic group over the field k , and $\mathfrak{g}$ its Lie algebra. Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$. Let G act on $W(\mathfrak{g})$ by the adjoint representation, and consider the subscheme $W(\mathfrak{h})$. The construction of no. 1 leads us to introduce

$$N = \text{Norm}_G(\mathfrak{h}) = \text{Norm}_G(W(\mathfrak{h})),$$

which is an algebraic subgroup of G (not necessarily smooth),

$$\mathfrak{n} = \text{Lie}(N),$$

and the scheme

$$X = G \times^N W(\mathfrak{h})$$

the quotient of $X = G \times W(\mathfrak{h})$ by N acting on the right by $(g, x) \cdot n = (gn, \text{Ad}(n^{-1})x)$. We consider the canonical morphisms

$$\begin{array}{ccc} G \times W(\mathfrak{h}) & & \\ \downarrow q & \searrow \varphi & \\ X & \xrightarrow{\psi} & W(\mathfrak{g}). \end{array}$$

Theorem 5.1. *With the preceding notation, suppose k infinite. Consider the following conditions:*

- (i) $\mathfrak{h}$ contains a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$.
- (ii) There exists $a \in \mathfrak{h}$ such that $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective.
- (iii) $\varphi : G \times W(\mathfrak{h}) \longrightarrow W(\mathfrak{g})$ is generically smooth.

⁽¹⁰⁾**N.D.E.** “The equality” has been replaced by “the equalities”.

- (iv) The preceding morphism φ (or again $\psi : X \rightarrow W(\mathfrak{g})$) is dominant, and $\mathfrak{h}$ has the same nilpotent rank as $\mathfrak{g}$.
- (v) $\psi : X \rightarrow W(\mathfrak{g})$ is generically smooth and $\mathfrak{h} = \mathfrak{n}$.
- (vi) $\psi : X \rightarrow W(\mathfrak{g})$ is dominant, and $\mathfrak{h} = \mathfrak{n}$.
- (vii) $\psi : X \rightarrow W(\mathfrak{g})$ is dominant.
- (viii) $\psi : X \rightarrow W(\mathfrak{g})$ is dominant, and N is smooth.
- (ix) $\psi : X \rightarrow W(\mathfrak{g})$ is dominant, and $\mathfrak{h} = \mathfrak{n}$, and $\mathfrak{h}$ is the Lie algebra of a smooth algebraic subgroup H of G .
- (x) $\psi : X \rightarrow W(\mathfrak{g})$ is generically quasi-finite, and $\mathfrak{n} = \mathfrak{h}$.
- (xi) $\psi : X \rightarrow W(\mathfrak{g})$ is generically étale, and $\mathfrak{n} = \mathfrak{h}$.
- (xii) There exists a smooth algebraic subgroup H of G with Lie algebra $\mathfrak{h}$, and $\mathfrak{h}$ contains a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$.
- (xiii) There exists $a \in \mathfrak{h}$ such that N and the transporter M_a of a into $\mathfrak{h}$ (cf. no. 1) coincide in a neighborhood of e , and $\mathfrak{n} = \mathfrak{h}$.

One then has the following diagram of implications:

$$\begin{array}{ccccccc}
 \text{(xi)} & \Longleftrightarrow & \text{(xii)} & \Leftrightarrow & \text{(xiii)} & \Rightarrow & \text{(viii)} \Leftrightarrow \text{(ix)} \Leftrightarrow \text{(x)} \\
 & & \Downarrow & & & & \Downarrow \\
 \text{(i)} & \Leftrightarrow & \text{(ii)} & \Leftrightarrow & \text{(iii)} & \Leftrightarrow & \text{(iv)} \Rightarrow \text{(v)} \Rightarrow \text{(vi)} \Rightarrow \text{(vii)}.
 \end{array}$$

When k has characteristic zero, all the conditions considered are equivalent. Finally, one has

$$\text{(xi)} \Leftrightarrow [\text{(i) and (viii)}] \Leftrightarrow [\text{(v) and (viii)}].$$

PROOF. Note first the trivial implications:

$$\text{(v)} \Rightarrow \text{(vi)} \Rightarrow \text{(vii)}, \quad \text{(ix)} \Rightarrow \text{(vi)}, \quad \text{(xi)} \Leftrightarrow [\text{(x) and (v)}].$$

Let us prove the equivalence of conditions (i) to (iv) and the fact that they imply (v). Implication (i) $\Rightarrow$ (ii) is trivial. On the other hand, (iii) means, when k is algebraically closed, that there exists a point of $G \times W(\mathfrak{h})$ rational over k at which the tangent map to φ is surjective, and one sees at once that one can take this point of the form (e, a) , where $a \in \mathfrak{h}$ (transforming it, if necessary, by an operation of $G(k)$). One concludes that if k is infinite (not necessarily algebraically closed) this condition (clearly sufficient) of generic smoothness is still necessary. Now the tangent map is easily calculated: identifying the tangent space to $W(\mathfrak{h})$ at a with $\mathfrak{h}$, it is the map

$$(\xi, x) \mapsto [\xi, a] + x$$

from $\mathfrak{g} \times \mathfrak{h}$ to $\mathfrak{g}$. Saying that it is surjective also means that $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is surjective, or, what amounts to the same thing, injective. This proves the equivalence of conditions (ii) and (iii). Moreover, (ii) clearly implies $\mathfrak{h} = \mathfrak{n}$, and (iii) implies that ψ is generically smooth, for if φ is smooth at a point u , it follows (q being flat) that ψ is smooth at $q(u)$. Thus (ii), (iii) imply (v). Let us prove that they imply (i). For this, note that since ψ is dominant, and the set of regular points of $\mathfrak{g}$ is dense open, it follows that $\mathfrak{h}$ contains regular elements of $\mathfrak{g}$, so that a “sufficiently general” element b of $\mathfrak{h}$ is regular in $\mathfrak{g}$ and satisfies $\text{ad}(b)_{\mathfrak{g}/\mathfrak{h}}$ injective, hence $\text{Nil}(b, \mathfrak{g}) \subset \mathfrak{h}$, so $\mathfrak{h}$ contains the Cartan subalgebra $\mathfrak{d} = \text{Nil}(b, \mathfrak{g})$. Thus (i), (ii), (iii) are equivalent; finally (i) $\Leftrightarrow$ (iv), for one has already noted that if $\mathfrak{h}$ contains a Cartan subalgebra, it has the same rank as $\mathfrak{g}$ (4.6), hence (i) $\Rightarrow$ (iv); conversely, if (iv) holds, then $\mathfrak{h}$ contains a regular element of $\mathfrak{g}$, and since it has the same rank as $\mathfrak{g}$, it contains a Cartan subalgebra by 4.6.

Let us prove the equivalence of conditions (viii) to (x). Note first the following facts:

Lemma 5.2. *a) If N is smooth, then*

$$\dim X \leq \dim G,$$

equality holding if and only if $\mathfrak{h} = \mathfrak{n}$.

b) If $\mathfrak{h} = \mathfrak{n}$, then

$$\dim X \geq \dim G,$$

equality holding if and only if N is smooth.

These assertions follow at once from the formula

$$\dim X = \dim G - \dim N + \dim_k \mathfrak{h},$$

and from the fact that $\dim N = \dim_k \mathfrak{n}$ is equivalent to N smooth.

This being established, (viii) $\Rightarrow$ (x), for (viii) implies $\dim X \geq \dim G$, hence, by 5.2 a), equality of these dimensions and $\mathfrak{h} = \mathfrak{n}$, hence (x), and one similarly sees (x) $\Rightarrow$ (viii) by applying 5.2 b). On the other hand, (viii) implies (ix), for it implies $\mathfrak{h} = \mathfrak{n}$, so $\mathfrak{h}$ is the Lie algebra of the smooth algebraic subgroup N of G , and conversely (ix) $\Rightarrow$ (viii), for since H normalizes its Lie algebra $\mathfrak{h}$, and it is contained in N , and since H is smooth and has the same Lie algebra as N , it follows that N is smooth.

Finally, let us prove the equivalence of conditions (xi), (xii), (xiii) and the fact that they imply (iii) (which will finish establishing our diagram of implications). One has (xi) $\Leftrightarrow$ (xiii), for if $\mathfrak{n} = \mathfrak{h}$, then by 5.2 b) one has $\dim X \geq \dim G$, hence (xi) is then equivalent (taking into account that $W(\mathfrak{g})$ is normal) to the fact that ψ is generically unramified, which is also equivalent to (xiii) by (1.1 (ii) $\Leftrightarrow$ (iii)), proceeding as above for the proof of (ii) $\Leftrightarrow$ (iii). Since (xi) $\Rightarrow$ (x) $\Rightarrow$ (viii) by what we have already seen, one sees that (xi) implies that N is smooth, i.e. $q : G \times W(\mathfrak{h}) \rightarrow X$ is smooth, so the composite $\varphi = \psi \circ q$ is generically smooth, i.e. one has (iii). Since (iii) $\Rightarrow$ (i), it also follows that (xi) $\Rightarrow$ (xii)⁽¹¹⁾. Finally (xii) $\Rightarrow$ (xi), for one clearly has (xii) $\Rightarrow$ (i), so, since one has seen (i) $\Rightarrow$ (iii) $\Rightarrow$ (v), one has (xii) $\Rightarrow$ (v); it follows that one also has (xii) $\Rightarrow$ (ix), and since one has seen (ix) $\Rightarrow$ (x), it follows that (xii) $\Rightarrow$ ((v) and (x)), hence (xii) $\Rightarrow$ (xi), since generically étale = generically smooth + generically quasi-finite.

Finally, when k has characteristic 0, then (vii) $\Rightarrow$ (viii), for by a theorem of Cartier, N is automatically smooth (VI_B 1.6.1.), and [(viii) and (x)] $\Rightarrow$ (xi), since in characteristic zero, for a morphism of integral preschemes, generically étale = dominant and generically quasi-finite. This shows that in this case, all conditions (i) to (xiii) are equivalent. $\square$

Corollary 5.3. *Under the equivalent conditions (viii) to (x), there exists a unique smooth connected algebraic subgroup H of G whose Lie algebra is $\mathfrak{h}$, and one has*

$$\text{Norm}_G(H) = \text{Norm}_G(\mathfrak{h}) = N, \quad H = N^0.$$

Indeed, $H = N^0$ will satisfy the desired conditions; on the other hand, if H satisfies them, then (since H normalizes its Lie algebra $\mathfrak{h}$) one has $H \subset N$, so, since this is an inclusion of smooth groups with the same Lie algebra, with H connected, one will have $H = N^0$. For the identity $\text{Norm}_G(H) = \text{Norm}_G(\mathfrak{h})$, one may suppose k algebraically closed; then from what one has just seen it follows immediately that the points of the two groups with values in k are the same; on the other hand the inclusions $H \subset \text{Norm}_G(H) \subset N$ show that $\text{Norm}_G(H)$ and N have the same Lie algebra, hence they are identical.

Corollary 5.4. *Under the equivalent conditions (i) to (iv), let $a \in \mathfrak{h}$. Then the following conditions are equivalent, and hold if a is regular in $\mathfrak{g}$:*

- (i) φ is smooth at (e, a) .
- (ii) $M_a = \text{Transp}_G(a, \mathfrak{h})$ is smooth at e , and $\dim_e(M_a) = \text{rank}_k \mathfrak{h}$.
- (iii) $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective (or again, bijective).

When one is under the equivalent conditions (xi) to (xiii), let H be the algebraic subgroup of G considered in 5.3. Then the preceding conditions are also equivalent to the following conditions:

⁽¹¹⁾**N.D.E.** For $H = N$ will do.

- (iv) ψ is étale at $(\bar{e}, a)$.
- (v) Denoting by M_a^0 the connected component of e in the transporter M_a of a into $\mathfrak{h}$, equipped with the structure induced by M_a , one has

$$H = M_a^0.$$

Clearly (i) $\Rightarrow$ (ii), since M_a is isomorphic to the fiber $\varphi^{-1}(a)$, the point e corresponding to (e, a) , and one has (ii) $\Rightarrow$ (i), for (ii) implies that φ is “equidimensional” at (e, a) (i.e. the dimension of the fiber through this point is that of the generic fiber), which implies ($G \times W(\mathfrak{h})$ and $W(\mathfrak{g})$ being regular) that it is flat at (e, a) , hence smooth since its fiber is so at this point. The equivalence of (i) and (iii) was seen in the proof of 5.1 as resulting from the simple calculation of the tangent map. Moreover, one has seen in 4.8 b) that “ a regular in $\mathfrak{g}$ ” $\Rightarrow$ (iii). Under conditions (xi) to (xiii), since $q : G \times W(\mathfrak{h}) \rightarrow X$ is smooth (N being smooth), it follows that (i) is equivalent to ψ smooth at $(\bar{e}, a)$, and since ψ is generically étale, this is equivalent to (iv). Finally, as was noted at the end of the proof of 1.1, (iv) implies that N is the prescheme induced on M_a by an open and closed part of M_a , whence (v); finally (v) $\Rightarrow$ (ii) trivially (or again (v) $\Rightarrow$ (iv) by 1.1, for ψ being dominant and $W(\mathfrak{g})$ normal, “unramified” is here equivalent to “étale”); this completes the proof of 5.4.

Corollary 5.5. *Let G be a smooth algebraic group over a field k , H a smooth algebraic subgroup such that its Lie algebra $\mathfrak{h}$ contains (at least over a suitable extension of k) a Cartan subalgebra of the Lie algebra $\mathfrak{g}$ of G . Let K be a connected algebraic subgroup of G (not necessarily smooth), with Lie algebra $\mathfrak{k}$; suppose that $\mathfrak{k}$ contains a regular element a of $\mathfrak{g}$ (at least over a suitable extension of k). Then H contains K if and only if $\mathfrak{h}$ contains $\mathfrak{k}$.*

Indeed, by 5.4 (iii) $\Rightarrow$ (v) one has $H = M_a^0$ (N.B. of course, this relation being invariant under extension of the base field, it is valid without the hypothesis that the latter be infinite!); on the other hand $\mathfrak{k} \subset \mathfrak{h}$ clearly implies $K \subset M_a$, so, since K is connected, $K \subset M_a^0$, whence $K \subset H$. Q.E.D.

Corollary 5.6. *Let G, H be as in 5.5 and let K be an algebraic subgroup of G ; one supposes H connected and K smooth. Then K contains H if and only if $\mathfrak{k}$ contains $\mathfrak{h}$.*

Indeed, if $\mathfrak{k} \supset \mathfrak{h}$, then K satisfies the hypothesis considered in 5.5 for H ; on the other hand H clearly satisfies the condition considered for K in 5.5. The conclusion then follows from 5.5.

Corollary 5.7. *Let $\mathfrak{g}$ be the Lie algebra of a smooth algebraic group G over a field k . Then:*
a) Let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{g}$. For $\mathfrak{d}$ to be a Cartan subalgebra, it is necessary and sufficient that $\mathfrak{d}$ be nilpotent and equal to its own normalizer.
b) Let a be an element of $\mathfrak{g}$; for a to be regular, it is necessary and sufficient that $\text{Nil}(a, \mathfrak{g})$ be nilpotent.

Extending the base field, one may suppose k infinite. Taking 4.4 into account, one is reduced for a) to proving that if $\mathfrak{d}$ is nilpotent and contains an element a such that $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective, then $\mathfrak{d}$ is a Cartan subalgebra. Now by 5.1 (ii) $\Rightarrow$ (i), $\mathfrak{d}$ contains a Cartan subalgebra $\mathfrak{d}' = \text{Nil}(a, \mathfrak{g})$, and by 4.3 one concludes from the fact that $\mathfrak{d}$ is nilpotent that $\mathfrak{d} = \mathfrak{d}'$. To prove b), one notes that $\text{Nil}(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$ by a), hence a is regular.

Corollary 5.8. *Let G be a smooth algebraic group over a field k , H a smooth algebraic subgroup, and $\mathfrak{g}, \mathfrak{h}$ the Lie algebras; suppose that after a suitable extension of the base field, $\mathfrak{h}$ contains a Cartan subalgebra of $\mathfrak{g}$. Let $\mathfrak{d}$ be a subalgebra of $\mathfrak{h}$; then it is a Cartan subalgebra of $\mathfrak{h}$ if and only if it is a Cartan subalgebra of $\mathfrak{g}$.*

Taking 4.8 c) into account, it remains to show that if $\mathfrak{d}$ is a Cartan subalgebra of $\mathfrak{h}$, it is a Cartan subalgebra of $\mathfrak{g}$, and for this one is reduced to showing that $\mathfrak{d}$ contains an element a regular in $\mathfrak{g}$, supposing (as one may) k algebraically closed. But since there is a dense open

subset of $\mathfrak{h}$ consisting of regular points of $\mathfrak{g}$, our assertion follows from 5.1 (i) $\Rightarrow$ (vii) applied to $(\mathfrak{h}, \mathfrak{d})$.

6. Cartan subalgebras and subgroups of type (C), relative to a smooth algebraic group

For simplicity, we confine ourselves in the following theorem to the case of an algebraically closed base field (the case of an arbitrary base prescheme being treated in the next exposé):

Theorem 6.1. *Let G be a smooth algebraic group over an algebraically closed field k , and $\mathfrak{g}$ its Lie algebra. Then:*

- a) *The Cartan subalgebras of $\mathfrak{g}$ are conjugate.*
- b) *Let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. Then its normalizer N in G is smooth, and $D = N^0$ is the only smooth connected subgroup of G whose Lie algebra is $\mathfrak{d}$. One has*

$$\text{Norm}_G(\mathfrak{d}) = \text{Norm}_G(D) = N, \quad \text{hence } D = \text{Norm}_G(D)^0.$$

- c) *With $\mathfrak{d}$ as in b), set, as in no. 5, $X = G \times^N W(\mathfrak{d})$, and consider the canonical morphism*

$$\psi : X \longrightarrow W(\mathfrak{g}),$$

(whose fiber at $a \in \mathfrak{g}$ has as its points with values in k the set of Cartan subalgebras of $\mathfrak{g}$ which contain a). Then ψ is a birational morphism.

- d) *With the notation of c), let U be the largest open subset of $W(\mathfrak{g})$ such that ψ induces an isomorphism*

$$\psi^{-1}(U) \xrightarrow{\sim} U.$$

Then for $a \in \mathfrak{g}$, the following conditions are equivalent:

- (i) $a \in U(k)$.
- (ii) a is contained in one and only one Cartan subalgebra of $\mathfrak{g}$.
- (iii) The set of Cartan subalgebras of $\mathfrak{g}$ containing a is finite and nonempty.
- (iv) The fiber $\psi^{-1}(a)$ has an isolated point.
- (v) (If $a \in \mathfrak{d}$) The morphism ψ is étale (or merely: quasi-finite) at the point $(\bar{e}, a)$.
- (vi) a is a regular element of $\mathfrak{g}$.
- (vii) With the notation of 5.4 (iii), one has $N = M_a$.

PROOF. Let us apply 5.1 when $\mathfrak{h}$ is a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$; let us show that the strongest conditions (xi) to (xiii) then hold. This is clear, either in form (xiii), taking 4.7 into account (which implies that for a regular element a of $\mathfrak{g}$ contained in $\mathfrak{d}$, one has $N = M_a$), or in form (xi) = (x) + (i), for condition (i) is trivial and condition (x) follows from the fact that a regular point of $\mathfrak{g}$ is contained in only one Cartan subalgebra of $\mathfrak{g}$ (4.6 a)), and a fortiori in only one conjugate of $\mathfrak{d}$. Then b) follows from 5.3, and c) follows from the fact that ψ is generically étale and that a sufficiently general point (more precisely, a regular point) of $\mathfrak{g}$ is contained in only one Cartan subalgebra of $\mathfrak{g}$. Under these conditions, the equivalence of conditions (i) to (v) on a is an immediate consequence of Zariski's Main Theorem for the separated birational morphism ψ , taking into account that $W(\mathfrak{g})$ is normal and X integral. The equivalence of (v) and (vi) is a particular case of 5.4 (iii) $\Leftrightarrow$ (iv) (reducing to the case where $a \in \mathfrak{d}$ by transforming a by a suitable element $g \in G(k)$), taking 4.6 b) into account. Moreover, by 5.4, (v) and (vi) are also equivalent to $N \supset M_a^0$, and by 4.7, already invoked, this even implies $N = M$. This proves d).

Of course, b), c), and the equivalence of (i), (iv), (v), (vi), (vii) remain valid over an arbitrary base field. Let us now prove a), using the fact that k is algebraically closed. By 5.1 (i) $\Rightarrow$ (vii), $\psi : X \longrightarrow W(\mathfrak{g})$ is dominant, hence there exists a dense open subset V of $W(\mathfrak{g})$ such that every $a \in V(k)$ is the image of an element of $X(k)$, i.e. is contained in a conjugate of $\mathfrak{d}$. Applying this result to a second Cartan subalgebra $\mathfrak{d}'$ of $\mathfrak{g}$, one sees that one may take V such that every element of $V(k)$ is conjugate to an element of $\mathfrak{d}$ and to an element of $\mathfrak{d}'$. Taking a regular element in $V(k)$, it follows that there is a conjugate $\mathfrak{d}''$ of $\mathfrak{d}'$ which contains

an element of $\mathfrak{d}$ regular in $\mathfrak{g}$, and hence is equal to $\mathfrak{d}$ by 4.6 a). This completes the proof of 6.1. $\square$

Definition 6.2. Let G be a smooth algebraic group over a field k . One calls a *subgroup of type (C)* of G any smooth connected algebraic subgroup whose Lie algebra is a Cartan subalgebra of $\mathfrak{g}$. One calls the *infinitesimal rank* of G the nilpotent rank of its Lie algebra $\mathfrak{g}$, equal to the dimension of every subgroup of type (C) of G .

From 6.1 b) one concludes at once:

Corollary 6.3. *Under the conditions of 6.2, the map $D \mapsto \mathfrak{d} = \text{Lie } D$ establishes a one-to-one correspondence between subgroups of type (C) of G and Cartan subalgebras $\mathfrak{d}$ of $\mathfrak{g}$. If D is a subgroup of type (C) of G , D is its own connected normalizer:*

$$D = \text{Norm}_G(D)^0.$$

Combining 6.1 (a) and 6.3, one finds:

Corollary 6.4. *If k is algebraically closed, the subgroups of type (C) of G are conjugate to one another.*

Corollary 6.5. *Let G be a smooth algebraic group over algebraically closed k , H a smooth algebraic subgroup of G , and $\mathfrak{g}$ and $\mathfrak{h}$ the Lie algebras. For $\mathfrak{h}$ to contain a Cartan subalgebra of $\mathfrak{g}$, it is necessary and sufficient that H contain a subgroup D of type (C) of G .*

This is clearly sufficient, and it is also necessary, for $H \supset D$ holds if and only if $\mathfrak{h} \supset \mathfrak{d}$, by 5.5.

Remarks 6.6. a) One will note that the connected subgroups H of G described in 6.5 correspond one-to-one to the Lie subalgebras $\mathfrak{h}$ of $\mathfrak{g}$ which satisfy the strongest condition (xii) of 5.1 (and by 5.5, the inclusion relations between such subgroups can already be recognized on their Lie algebras).

b) Still suppose k algebraically closed, and let D be a subgroup of type (C) of G ; then it is easy to show that D contains a Cartan subgroup C of G : indeed, let $V = \mathfrak{g}/\mathfrak{d}$; then for a “general” element a of $\mathfrak{d}$, $\text{ad}(a)$ acting on V is injective (it suffices that a be regular in $\mathfrak{g}$), whence one easily concludes that for “general” $g \in D(k)$, $\text{Ad}(g)$ acting on V has no nonzero invariant, which allows one to apply 2.1 (vii) $\Rightarrow$ (i). In fact, we shall see in the next exposé a more precise result: every Cartan subgroup C of G is contained in one and only one subgroup of type (C) of G .

c) One should take care not to confuse Cartan subgroups and subgroups of type (C) in general: the subgroups of type (C) of G are Cartan subgroups if and only if they are nilpotent (Cartan subgroups being indeed maximal nilpotent subgroups); now it may happen that $\mathfrak{g}$ is nilpotent without G being so (example: $\text{SL}(2)_k$ for k of characteristic 2); then the Cartan subalgebras of $\mathfrak{g}$ are identical to $\mathfrak{g}$, i.e. G is a subgroup of type (C) of itself if G is connected, but it is not a Cartan subgroup of G ! On the other hand, we shall see in XIV that if G is an adjoint semisimple group, then its subgroups of type (C) are its Cartan subgroups. Likewise, in characteristic 0, without restriction on the smooth algebraic group G over k , the same conclusion holds: this follows at once from the fact that in characteristic 0, a connected algebraic group is nilpotent if (and only if) its Lie algebra is. This follows at once from the fact that in characteristic zero, the center Z of the connected algebraic group G has as Lie algebra the center of $\mathfrak{g}$ (for one obtains a priori the space of invariants of $\mathfrak{g}$ under the adjoint representation of G ; now G being connected and k of characteristic zero, these are also the “invariants” in the infinitesimal sense of the adjoint representation of $\mathfrak{g}$, i.e. the center of $\mathfrak{g}$), and that by Cartier’s theorem (cf. VI_B 1.6.1) Z is smooth.